

Raghavendra Pedada

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Final-year Computer Science undergraduate (IoT, Cybersecurity & Blockchain; CGPA 8.03; graduating 2027) with a strong foundation in data structures, algorithms, OOP, operating systems and DBMS. Built 5+ end-to-end systems in Python, C++, JavaScript, React and FastAPI, including distributed multi-tier pipelines (device → broker → dashboard) and on-device ML tuned for tight latency and CPU budgets. Works across two tracks, IoT/embedded systems and AI/software development, with edge AI (on-device inference, no cloud dependency) as a core strength. Skilled at decomposing broadly defined problems, writing clean, maintainable code and clearly explaining design trade-offs.

Education

Bachelor of Technology, Computer Science & Engineering Expected: 2027

Alva's Institute of Engineering and Technology (VTU), Mangaluru, Karnataka

CGPA: 8.03

Specialization: IoT, Cybersecurity & Blockchain

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Cryptography & Network Security, Embedded Systems Design, ARM Architecture, Color Image Processing

Intermediate (Class XII)

07/2023

Narayana Junior College, Andhra Pradesh

Score: 78%

Secondary School (Class X)

07/2021

Sri Chaithanya Techno School, Andhra Pradesh

Score: 99%

Technical Skills

Programming Languages: Python, C++, Java, C, JavaScript, SQL

Core Computer Science: Data Structures & Algorithms, Complexity Analysis, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks

Backend & Web: FastAPI, REST APIs, React, Vite, HTML5, CSS3, SQLAlchemy, Axios, Chart.js, Framer Motion

Databases: SQLite, MySQL

IoT & Embedded Systems: MQTT, Node-RED, HiveMQ Cloud, Raspberry Pi, ESP32, Arduino, Multi-tier Architectures, Sensor Integration, Linux

AI/ML & Edge AI: TensorFlow Lite, MediaPipe, OpenCV, RAG (Retrieval-Augmented Generation), LLM Function Calling, Prompt Engineering, NumPy, pandas

DevOps & Tools: Git, GitHub, Docker, GitHub Actions, CI/CD (OpenShift, Tekton), VS Code, MATLAB

Experience

Software Intern

Sep 2026 – Present

IFB Automotive Private Limited

Bangalore, Karnataka (In-office)

- Working on IFB Connect, an internal Windows messaging application that runs entirely within the company's local network (LAN-only), with a modern glassmorphism user interface.
- Studied the end-to-end product flow from incoming materials to final shipment on the factory floor and maintain structured daily technical reports.

Cybersecurity Micro-Intern

2024

Tata Consultancy Services (TCS)

Virtual Programme

- Covered threat modelling and vulnerability assessment.
- Evaluated IoT security postures and proposed mitigations.

Projects

Edge-AI Smart Mirror (Final Year Project) | *Raspberry Pi 4B, TensorFlow Lite, MediaPipe, OpenCV, Python*

- Built a real-time, on-device health and wellness monitor on a Raspberry Pi 4B (ARM Cortex-A72) with zero cloud or internet dependency, shown as a live overlay on a two-way mirror.
- Implemented mood detection (7-class FER-2013 model mapped to 4 wellness states, 93% accuracy) and posture monitoring (MediaPipe BlazePose, 33 landmarks, 89% accuracy) with <50 ms inference latency.
- Designed a modular OpenCV vision architecture for concurrent multi-metric tracking using OOP and clean-code principles.
- Led design decisions for a 4-member team, presented to a faculty panel, and co-authored an IEEE conference paper on the system (submitted).

HashPilot – Computational Benchmark Platform | *FastAPI, React, SQLite, Docker, GitHub Actions*

- Built a full-stack platform benchmarking Proof-of-Work strategies (Sequential, Random, Multi-Thread, Multi-Process) to analyze performance and concurrency trade-offs.
- Developed a FastAPI backend with REST APIs for benchmark execution, analytics and AI-powered strategy recommendations, plus an interactive React dashboard with performance charts.
- Added SQLite persistence, Docker containerization and GitHub Actions CI/CD, owning the full development lifecycle.

IoT Health Monitoring System | *ESP32, MQTT, Python, Sensors*

- Built a multi-tier pipeline streaming real-time biometric sensor data from ESP32 devices over MQTT to a Python backend, bridging hardware, messaging and application layers.

AQI Monitoring Station (Simulated IoT System) | *Python, MQTT, Node-RED, HiveMQ Cloud*

- Built a full Air Quality Index monitoring station as a simulated IoT project, with a Python MQTT simulator publishing through a HiveMQ Cloud broker to a Node-RED dashboard.

Cosmic RAG System & AI Tool Calling Assistant | *Python, RAG, LLMs, Function Calling*

- Built a RAG pipeline with separate indexing, semantic retrieval and generation layers for document question answering.
- Developed an assistant that invokes external tools via structured function calling to automate multi-step API workflows.

Invisible Patient – Hospital Ward Communication System | *Cognizant Technoverse Hackathon 2026 (Participant)*

- Built a working prototype for government hospitals (QR wristbands, live agent-powered dashboard) under hackathon time limits.

3D Personal Portfolio Website | *React, Vercel*

- Designed and deployed a space-themed, React-based 3D portfolio website, live at raghavendrap.vercel.app.

Foundational IoT Projects | *Arduino, ESP32, Raspberry Pi, MQTT*

- Built a Smart Home Automation System and a Smart Waste Management System using Arduino, ESP32, Raspberry Pi and MQTT.

Certifications & Publications

- **IBM / Coursera:** CI/CD with OpenShift: GitHub Actions & Tekton Pipelines (2026)
- **IEEE Conference Paper (Co-author, Submitted):** Edge-AI Smart Mirror: Hybrid Mood & Posture Detection System